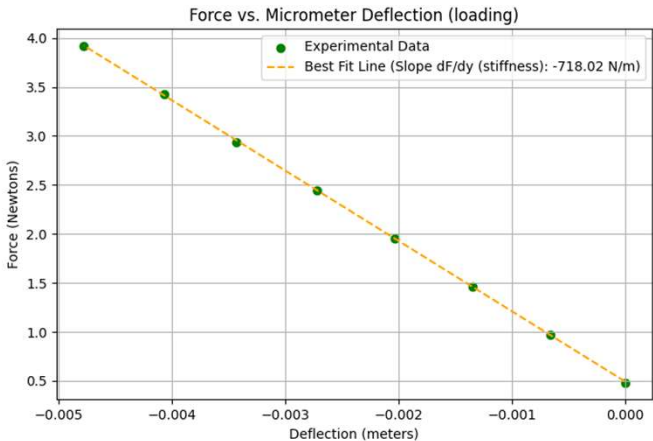


Zeshui Song, 5/4/26, HW10

10.0 (For the lab report) Derive an equation that relates the force vs. deflection data slope and beam dimensions to Young's modulus. Use this deflection equation to estimate Young's modulus of the beam material.

Best fit slope of force vs deflection

- Using python, I will input the experimental data and plot a best fit line. Converting from cumulative load (g) to force (N) and micrometer readings to deflection using: (deflection) = (current - initial)
- This will result in a negative deflection in the y direction as load increases



1

Zeshui Song, 5/4/26, HW10

Deriving an equation for the stiffness (or dF/dy_{micro})

Approach:

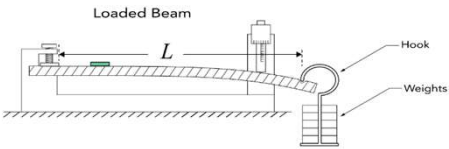
- Using a FBD to determine the moment distribution $M(x)$ starting from the clamp location ($x=0$)
- Solve ODE using initial conditions (fixed support):

$$\left. \frac{dy}{dx} \right|_{x=0} = 0, \quad y(0) = 0$$
$$\frac{d^2y}{dx^2} = \frac{M(x)}{EI}$$

- Integrating ODE using initial conditions:

$$\frac{dy}{dx} = \int_0^x \frac{M(x')}{EI} dx' + \left. \frac{dy}{dx} \right|_{x=0} = \int_0^x \frac{M(x')}{EI} dx' + 0$$
$$y = \int_0^x \int_0^{x'} \frac{M(x'')}{EI} dx'' + y(0)$$

- Find $y_{micro} = y(x_{c \rightarrow l} - x_{l \rightarrow m})$, the *deflection at the micrometer measuring point*. (Subtract because the micrometer is to the left of the load)
- $M(x)$ is in terms of the load F , thus rewriting the resulting deflection equation for $F(y_{micro})$ and differentiating for dF/dy_{micro} will allow me to solve for young's modulus



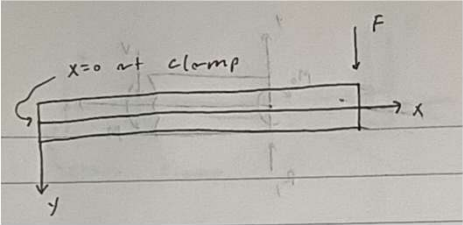
2

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Geometry

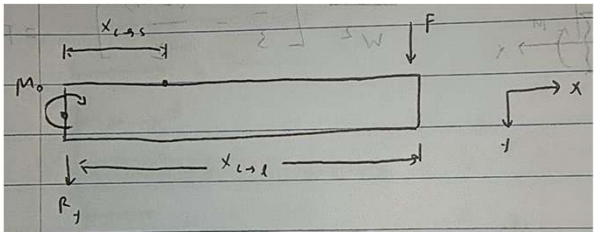
Given:

- $t = 0.1260$ in (Thickness)
- $L = 12.5$ in (Length)
- $w = 1.0020$ in (Width)
- Clamp to loading point $x_{c \rightarrow l} = 11.515$ in
- Loading point to micrometer $x_{l \rightarrow m} = 1.468$ in
- Clamp to strain gauge center $x_{c \rightarrow s} = 0.343$ in



$$I = \frac{1}{12}bh^3 = \frac{1}{12}wt^3$$

FBD to determine stress distribution



$$\sum F_y = 0 = R_y + F \rightarrow R_y = -F$$

$$\sum M_z = 0 = M_0 + Fx_{c \rightarrow l} \rightarrow M_0 = -Fx_{c \rightarrow l}$$

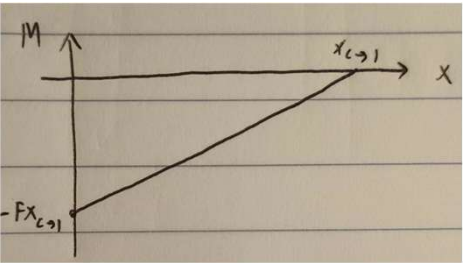
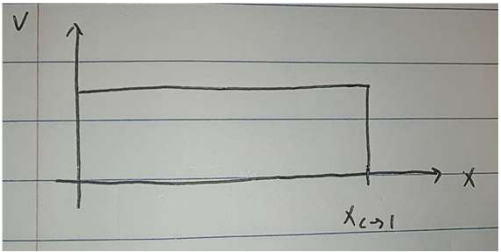
3

Zeshui Song, 5/4/26, HW10

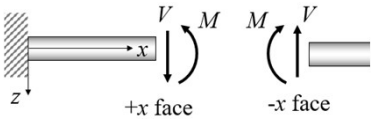
Starting from the clamp, using the differential form:

$$\frac{dV}{dx} = -w \rightarrow V = V(0) = F$$

$$\frac{dM}{dx} = V \rightarrow M = M(0) + \int_0^x V dx' \rightarrow M = -Fx_{c \rightarrow l} + Fx$$



Note: sign convention



4

Zeshui Song, 5/4/26, HW10

Solve for deflection y

$$\frac{dy}{dx} = \int_0^x \frac{-Fx_{c \rightarrow l} + Fx'}{EI} dx' + 0 = \frac{F}{EI} \left[-x_{c \rightarrow l}x + \frac{x^2}{2} \right]$$

$$y = \int_0^x \frac{F}{EI} \left[-x_{c \rightarrow l}x' + \frac{x'^2}{2} \right] dx' + 0 = \frac{F}{EI} \left[\frac{-x_{c \rightarrow l}x^2}{2} + \frac{x^3}{6} \right]$$

Solve for deflection y_{micro}

$$y_{micro} = y(x_{c \rightarrow l} - x_{l \rightarrow m})$$

$$\text{Let } B = x_{c \rightarrow l} - x_{l \rightarrow m}$$

$$y_{micro} = \frac{F}{EI} \left[\frac{-x_{c \rightarrow l}B^2}{2} + \frac{B^3}{6} \right]$$

Solve for stiffness (or dF/dy_{micro})

$$F = y_{micro}EI \left[\frac{-x_{c \rightarrow l}B^2}{2} + \frac{B^3}{6} \right]^{-1}$$

$$\frac{dF}{dy_{micro}} = EI \left[\frac{-x_{c \rightarrow l}B^2}{2} + \frac{B^3}{6} \right]^{-1}$$

5

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Rewrite to solve for young's modulus

$$E = \frac{\left(\frac{dF}{dy_{micro}} \right)}{I} \left[\frac{-x_{c \rightarrow l}B^2}{2} + \frac{B^3}{6} \right]$$

$$E = \frac{\left(\frac{dF}{dy_{micro}} \right)}{\frac{1}{12}wt^3} \left[\frac{-x_{c \rightarrow l}B^2}{2} + \frac{B^3}{6} \right] = \frac{12 \left(\frac{dF}{dy_{micro}} \right)}{wt^3} \left[\frac{-x_{c \rightarrow l}B^2}{2} + \frac{B^3}{6} \right]$$

Plugging in values

$$B = x_{c \rightarrow l} - x_{l \rightarrow m} = \left[11.515 - 1.468 \text{ in} \cdot \frac{0.0254 \text{ m}}{1 \text{ in}} \right] = 0.255 \text{ m}$$

$$E = \frac{12 \left[-718.02 \frac{N}{m} \right]}{\left[1.0020 \cdot 0.1260^3 \text{ in}^4 \cdot \left(\frac{0.0254 \text{ m}}{1 \text{ in}} \right)^4 \right]} \left[\frac{- \left[11.515 \text{ in} \cdot \frac{0.0254 \text{ m}}{1 \text{ in}} \right] [0.255 \text{ m}]^2}{2} + \frac{[0.255 \text{ m}]^3}{6} \right]$$

$$\boxed{E = 6.966 \times 10^{10} \text{ N/m}^2}$$

While the value for young's modulus I found using deflection didn't match the one I found using strain over the force ($d\varepsilon/dF$) in HW 9, this value of $E = 6.966 \times 10^{10} \text{ N/m}^2$ is very close to that modulus for Aluminum Alloy 1100-H14 (70 GPa). Thus, I think that the material is some alloy of aluminum and the discrepancy in calculations is due to measurement error most likely. [Time spent: 60 mins]

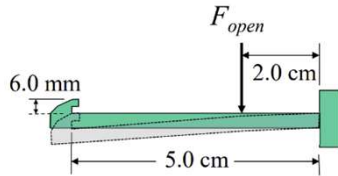
6

Zeshui Song, 5/4/26, HW10

10.1. (GRADED) Complete and check your in-class design problem. Write up your problem in the following format:

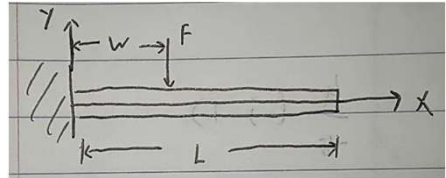
A. Problem description: describe your design goal and constraints

Given a door catch prong of $L=5$ cm, the tip should deflect 6.0 mm given a load of $F_{open} < 25$ N at 2 cm from the cantilevered end. Design the geometry of the prong cross section after selecting a material.



B. Describe the model used to analyze your design

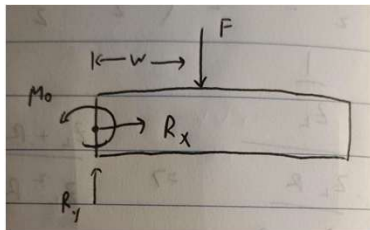
Deriving a moment distribution using a FBD of the prong, I can use the ODE $\frac{d^2y}{dx^2} = \frac{M(x)}{EI}$ to solve for the deflection y given initial conditions at the cantilevered end.



7

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FBD

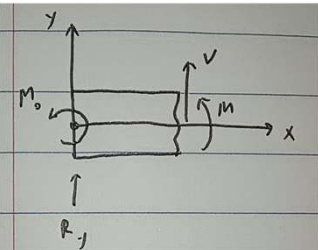


$$\sum F_x = 0 = R_x \rightarrow R_x = 0$$

$$\sum F_y = 0 = R_y - F \rightarrow R_y = F$$

$$\sum M_{z,0} = 0 = M_0 - Fw \rightarrow M_0 = Fw$$

For $0 \leq x \leq w$:



$$\sum F_y = 0 = R_y + V \rightarrow V = -R_y$$

$$\boxed{V = -F}$$

$$\sum M_{z,0} = 0 = M_0 + M + Vx \rightarrow M = -M_0 - Vx = -Fw + R_yx$$

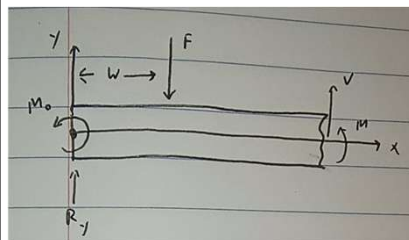
$$\boxed{M = -Fw + Fx}$$

NOTE: The differential equations for internal shear and bending moments ($\frac{dV}{dx} = -w$, $\frac{dM}{dx} = V$) is defined for the downwards positive convention! Since I defined it differently here, those ODEs won't match, there will be a sign conflict.

8

Zeshui Song, 5/4/26, HW10

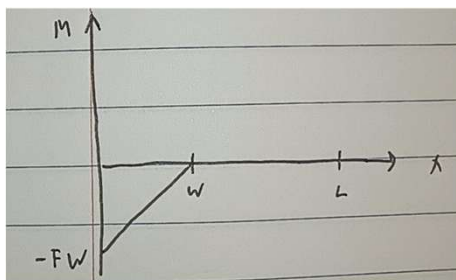
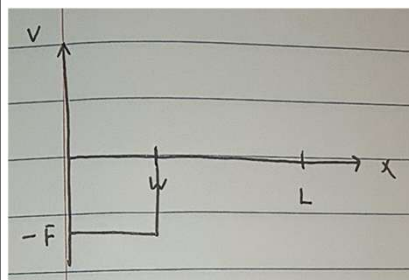
For $w \leq x \leq L$:



$$\sum F_y = 0 = R_y - F + V \rightarrow V = F - R_y \quad \boxed{V = 0}$$

$$\sum M_{z,0} = 0 = M_0 + M + Vx - Fw \rightarrow M = Fw - M_0 - Vx = Fw - Fw - 0 \quad \boxed{M = 0}$$

Stress and Moment Distribution:



9

Zeshui Song, 5/4/26, HW10

- To find the deflection at the tip of the cantilever, I will need use a tangent line from $w \leq x \leq L$ where $M=0$

$$y(x) = y(w) + y'(w) \cdot (x - w) \quad \text{for } (w \leq x \leq L)$$

Solving for the deflection using the initial conditions: $y(0) = 0$, $y'(0) = 0$

$$\frac{d^2 y}{dx^2} = \frac{M(x)}{EI} = \frac{1}{EI} [-Fw + Fx]$$

$$\frac{dy}{dx} = \int_0^x \frac{1}{EI} [-Fw + Fx] dx' + y'(0) = \frac{F}{EI} \left[-wx + \frac{x^2}{2} \right]$$

$$y = \int_0^x \frac{F}{EI} \left[-wx + \frac{x^2}{2} \right] dx' + y(0) = \frac{F}{EI} \left[\frac{-wx^2}{2} + \frac{x^3}{6} \right]$$

Thus, the MAX deflection at $x=L$ is:

$$y_{max} = y(w) + y'(w) \cdot (L - w) = \frac{F}{EI} \left[\frac{-w^3}{2} + \frac{w^3}{6} \right] + \frac{F}{EI} \left[-w^2 + \frac{w^2}{2} \right] (L - w)$$

$$\boxed{y_{max} = \frac{-Fw^3}{3EI} - \frac{FW^2}{2EI} (L - w)}$$

10

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C. Demonstrate using the model that your design (choice of material and dimensions) fulfils the design goal

Using the max deflection formula, I can now solve for the area moment of inertia, picking Nylon as the material ($E=2.8 \text{ GPa}$)

Given:

- $F = F_{open} < 25 \text{ N}$
- $E = 2.8 \times 10^9 \frac{\text{N}}{\text{m}^2}$
- $L = 0.05 \text{ m}$
- $w = 0.02 \text{ m}$
- $y_{max} = -0.006 \text{ m}$

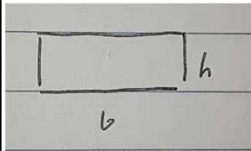
$$y_{max} = \frac{Fw^2}{EI} \left[\frac{-w}{3} - \frac{(L-w)}{2} \right]$$

$$\frac{y_{max}EI}{w^2} \left[\frac{-w}{3} - \frac{(L-w)}{2} \right]^{-1} = F < [25 \text{ N}]$$

$$I[-1.9384 \times 10^{12} \text{ N/m}^4] < [25 \text{ N}]$$

$$I < 1.289 \times 10^{-11} \text{ m}^4$$

Picking $I = 1.25 \times 10^{-11} \text{ m}^4$ and solving for the geometry of the cross section:



Choose $h = 3 \text{ mm}$

$$I = \frac{1}{12}bh^3$$

$$b = \frac{12I}{h^3} = \frac{12[1.25 \times 10^{-11} \text{ m}^4]}{[0.003 \text{ m}]^3} = 0.0055 \text{ m}$$

11

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Thus, Nylon with a rectangular cross section of (5.55 mm x 3 mm) satisfies the design criteria.

D. Discuss briefly the pros and cons of your design.

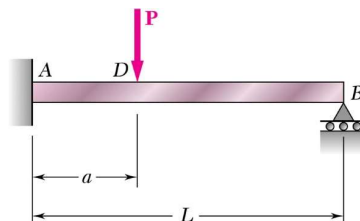
A thickness (h) of 3 mm keeps normal stresses low during the required 6 mm deflection, significantly reducing the risk of cyclic fatigue failure. Additionally, Nylon is a polymer that can stay within the linear elastic region without deforming. Some cons is that the design could be more efficient since the bending moment is zero past 2 cm from the support, thus material can be reduced near the tip.

[Time spent: 1 hr]

12

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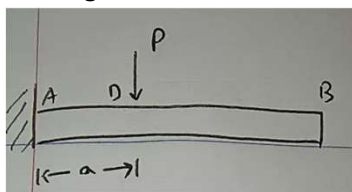
10.2. (GRADED) Beam AB has Young's modulus E , and moment of inertia I . Determine the reaction at the roller support and the deflection at point D located distance a from the rigid support, in terms of the problem parameters. Then give the results (deflection at D and load on support B) when a is equal to $L/2$.

**Approach**

- Considering the support at B as redundant and replacing it with its reaction R_B , yields two loadings
- Separating the two loadings using tabulated formulas in Appendix D will allow me to solve for the total deflection at B using superposition. Since B is a support, total deflection should be zero. Solving this will give me R_B in terms of P
- For deflection at D, I can use the superposition of the deflections at D from the two loadings
- Finally, I just need to substitute $a = L/2$ to get the new results

13

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Loading 1

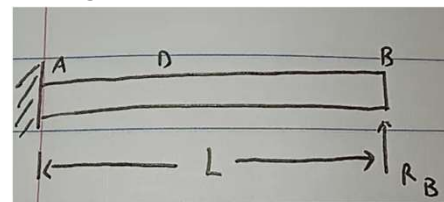
Section AD:

$$y_{AD,1}(x) = \frac{P}{6EI}(x^3 - 3ax^2) \longrightarrow y_{D,1} = \frac{P}{6EI}(a^3 - 3a^3) = \frac{-Pa^3}{3EI}$$

Section DB is straight since there is zero moment. The slope at D is:

$$\theta_{D,1} = -\frac{Pa^2}{2EI}$$

$$y_{DB,1}(x) = y_{D,1} + \theta_{D,1}(x - a) \longrightarrow y_{B,1} = \frac{-Pa^3}{3EI} - \frac{Pa^2}{2EI}(L - a)$$

Loading 2

Section AB:

$$y_{AB,2}(x) = \frac{-R_B}{6EI}(x^3 - 3Lx^2) \longrightarrow y_{D,2} = \frac{-R_B}{6EI}(a^3 - 3La^2)$$

$$y_{B,2} = \frac{R_B L^3}{3EI}$$

14

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Using the constraint at B to solve for the reaction at B in terms of P

$$\begin{aligned}
 y_{B,1} + y_{B,2} &= 0 \\
 \frac{-Pa^3}{3EI} - \frac{Pa^2}{2EI}(L-a) + \frac{R_B L^3}{3EI} &= 0 \\
 \frac{R_B L^3}{3EI} &= \frac{Pa^2}{EI} \left(\frac{a}{3} + \frac{1}{2}(L-a) \right) \\
 R_B &= \frac{3Pa^2}{L^3} \left(\frac{a}{3} + \frac{1}{2}(L-a) \right) \\
 \boxed{R_B = \frac{Pa^2}{2L^3}(3L-a)}
 \end{aligned}$$

Superposition from both loadings to find deflection at D

$$\begin{aligned}
 y_D &= y_{D,1} + y_{D,2} \\
 y_D &= \frac{-Pa^3}{3EI} - \frac{R_B}{6EI}(a^3 - 3La^2) \\
 y_D &= \frac{-Pa^3}{3EI} - \frac{1}{6EI} \frac{Pa^2}{2L^3}(3L-a)(a^3 - 3La^2) \\
 y_D &= \frac{-Pa^3}{3EI} + \frac{1}{6EI} \frac{Pa^2}{2L^3}a^2(3L-a)(3L-a) \\
 \boxed{y_D = -\frac{Pa^3}{3EI} \left(1 - \frac{a}{4L^3}(3L-a)^2 \right)}
 \end{aligned}$$

15

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Substituting $a = \frac{L}{2}$:

$$\begin{aligned}
 R_B &= \frac{P \left(\frac{L}{2} \right)^2}{2L^3} \left(3L - \frac{L}{2} \right) \\
 R_B &= \frac{PL^2}{8L^3} \left(\frac{5L}{2} \right) \\
 \boxed{R_B = \frac{5P}{16}}
 \end{aligned}$$

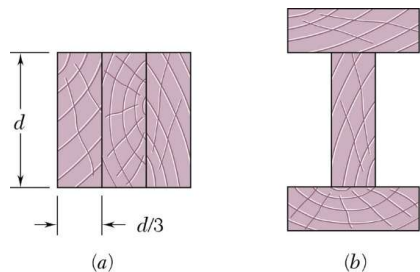
These results make sense because we get a positive R_B , which is consistent with the direction needed to oppose P. Also, the signs on the deflections make sense. The deflection from loading 2 will be upwards (+) while loading 1 will result in downwards deflection (-).
[Time spent: 35 mins]

$$\begin{aligned}
 y_D &= -\frac{P \left(\frac{L}{2} \right)^3}{3EI} \left(1 - \frac{L}{4L^3} \left(3L - \frac{L}{2} \right)^2 \right) \\
 y_D &= -\frac{PL^3}{24EI} \left(1 - \frac{L}{8L^3} \left(\frac{5L}{2} \right)^2 \right) \\
 y_D &= -\frac{PL^3}{24EI} \left(1 - \frac{L}{8L^3} \frac{25L^2}{4} \right) \\
 \boxed{y_D = -\frac{7PL^3}{768EI}}
 \end{aligned}$$

16

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10.4. A column of effective length L can be made by gluing together identical planks in either of the arrangements shown. Determine the ratio of the critical compression load using the arrangement a , to the critical load using the arrangement b .



Approach

- Using the formula for the critical load P for mode 1 buckling, for the same end conditions (equal L_e) and equal E :

$$P_{cr} = \frac{\pi^2 EI}{L_e^2}$$

- For each arrangement, the area moment of inertia is the only thing that changes, thus:

$$\frac{P_{cr,a}}{P_{cr,b}} = \frac{I_a}{I_b}$$

- Making sure to use the *smaller value of I* , since the critical load will tend to buckle the beam along its weaker axis.
 - For (a), this is where $b = d$, $h = d$. Both axes have equal I since it's a square.
 - For (b), this is along its Y-Y axis (vertical axis)

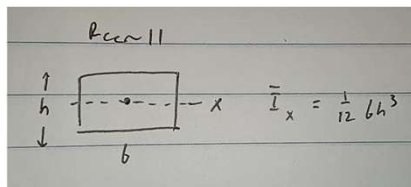
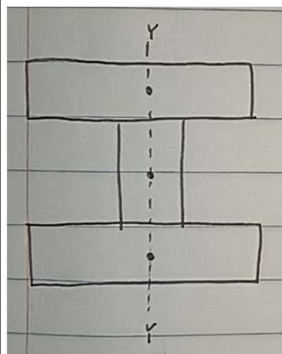
17

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Area moment of inertia for a:

$$I_a = \frac{1}{12}bh^3 = \frac{1}{12}d^4$$

Area moment of inertia for b:



$$I_b = \frac{1}{12}d\left(\frac{d}{3}\right)^3 + 2\left[\frac{1}{12}\left(\frac{d}{3}\right)d^3\right] = \frac{19}{324}d^4$$

Ratio:

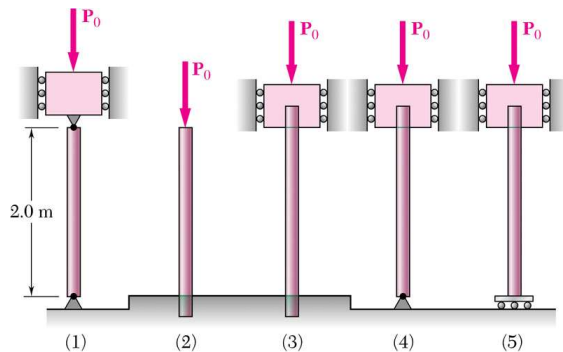
$$\frac{P_{cr,a}}{P_{cr,b}} = \frac{I_a}{I_b} = 1.421$$

Thus, the square beam is stronger under buckling loads because while the I beam is strong under bending in the X-X axis, it is weak in the Y-Y. [Time spent: 20 mins]

18

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10.5. (GRADED) Each of the five struts consists of an aluminum tube that has a 32-mm outer diameter and a 4-mm wall thickness. Using $E = 70 \text{ GPa}$ and a factor of safety of 2.3, determine the allowable load P_0 for each support condition shown. (Note the base in configuration (5) cannot rotate but can translate laterally. Hint: find the maximum wavelength allowed by the end conditions.)



Given:

- $d_o = 32 \text{ mm}$
- $t = 4 \text{ mm}$
- $E = 70 \text{ GPa}$
- $F.S. = 2.3$
- $L = 2 \text{ m}$

Find:

- Allowable load P_0 for each support condition

19

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Approach

- Using the formula for the critical load P for mode 1 buckling. The beams all have the same I, E , but varying L_e :

$$P_{cr} = \frac{\pi^2 EI}{L_e^2}$$

- The area moment of inertia of a hollow circular rod is given as:

$$I = \frac{\pi}{64} (d_o^4 - d_i^4), \quad d_i = d_o - 2t$$

- The effective lengths will be determined by the varying end conditions
- Using the factor of safety to find the allowable load

$$F.S. = \frac{P_{cr}}{P_{all}} \rightarrow P_{all} = P_0 = \frac{P_{cr}}{F.S.}$$

Geometry:

$$d_i = d_o - 2t = \left[32 - 2 \cdot 4 \text{ mm} \cdot \frac{10^{-3} \text{ m}}{1 \text{ mm}} \right] = 0.024 \text{ m}$$

$$I = \frac{\pi}{64} (d_o^4 - d_i^4) = \frac{\pi}{64} ([0.032 \text{ m}]^4 - [0.024 \text{ m}]^4) = 3.518 \times 10^{-8} \text{ m}^4$$

Allowable load in terms of only effective length:

$$P_{cr} = \frac{\pi^2 [70 \times 10^9 \text{ N/m}^2] [3.518 \times 10^{-8} \text{ m}^4]}{L_e^2} = \frac{[24304.88 \text{ Nm}^2]}{L_e^2}$$

$$P_0 = \frac{P_{cr}}{F.S.} = \frac{[10567.342 \text{ Nm}^2]}{L_e^2}$$

20

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(1) Both ends pinned: $L_e = L$

$$P_0 = \frac{[10567.342 \text{ Nm}^2]}{[2 \text{ m}]^2} = 2641.83 \text{ N}$$

(2) One fixed end, one free end: $L_e = 2L$

$$P_0 = \frac{[10567.342 \text{ Nm}^2]}{[2 \cdot 2 \text{ m}]^2} = 660.45 \text{ N}$$

(3) Both ends fixed: $L_e = 0.5L$

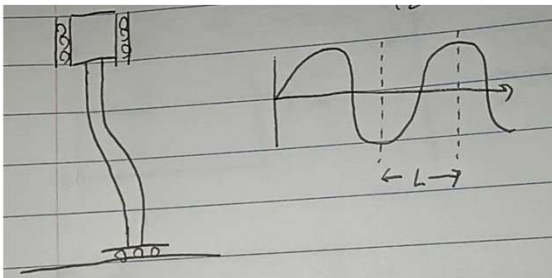
$$P_0 = \frac{[10567.342 \text{ Nm}^2]}{[0.5 \cdot 2 \text{ m}]^2} = 10567.342 \text{ N}$$

(4) One end fixed, one end pinned: $L_e = 0.7L$

$$P_0 = \frac{[10567.342 \text{ Nm}^2]}{[0.7 \cdot 2 \text{ m}]^2} = 5391.50 \text{ N}$$

21

Zeshui Song, 5/4/26, HW10

(5) One end fixed, one end free to translate but not rotate:

Since both ends cannot rotate, the slopes must be zero at the ends. Thus, the shape of bending is half of a full sine cycle and the wavelength is $2L$.

Based on the wavelength, the effective length is half of a full wavelength (based on the pinned-pinned beam definition of effective length):

$$L_e = \frac{\lambda}{2} = L$$

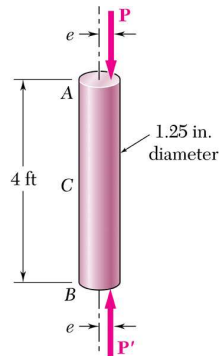
$$P_0 = \frac{[10567.342 \text{ Nm}^2]}{[2 \text{ m}]^2} = 2641.83 \text{ N}$$

These results show that the types of support/end conditions can affect the allowable load since it affects the buckling shape of the beam. [Time spent: 40 mins]

22

Zeshui Song, 5/4/26, HW10

10.6. An axial load P is applied to the 1.25-in diameter steel rod AB as shown. For $P = 8.6$ kips and $e = 1/16$ " determine (a) the deflection at the midpoint C of the rod, (b) the maximum stress in the rod. Use $E = 29 \times 10^6$ psi.



Given:

- $d = 1.25$ in
- $P = 8.6$ kips
- $e = \frac{1}{16}$ in
- $E = 29 \times 10^6$ psi
- $L = 4$ ft

Find:

- Deflection at C
- Max stress in the rod

23

Zeshui Song, 5/4/26, HW10

Approach

- The max deflection in the beam is at C . Thus, using the formula for eccentric loading:

$$y_{max} = y_D = e \left[\sec \left(\frac{\pi}{2} \sqrt{\frac{P}{P_{cr}}} \right) - 1 \right]$$

- The max stress in the rod is given by

$$\sigma_{max} = \frac{P}{A} \left[1 + \frac{ec}{r^2} \sec \left(\frac{\pi}{2} \sqrt{\frac{P}{P_{cr}}} \right) \right]$$

This equation is the superposition of the axial stress $\frac{P}{A}$ and the max bending stress $\frac{M_{max}c}{I}$ where $M_{max} = P(e + y_{max})$. (Note that $I = A \cdot r^2$, where r is the radius of gyration).

- Like before, the critical load is given by

$$P_{cr} = \frac{\pi^2 EI}{L_e^2}$$

Geometry:

$$I = \frac{\pi}{64} d^4 = \frac{\pi}{64} [1.25 \text{ in}]^4 = 0.1198 \text{ in}^4$$

$$L_e = L = 48 \text{ in (pinned - pinned case)}$$

$$c = \frac{d}{2} = 0.625 \text{ in}$$

$$r = \sqrt{\frac{I}{A}} = \sqrt{\frac{\frac{\pi}{64} d^4}{\frac{\pi}{4} d^2}} = \frac{d}{4} = 0.3125 \text{ in} \quad A = \frac{\pi}{4} d^2 = 1.22718 \text{ in}^2$$

24

Zeshui Song, 5/4/26, HW10

Solve for deflection:

$$P_{cr} = \frac{\pi^2 EI}{L_e^2} = \frac{\pi^2 [29 \times 10^6 \text{ psi}] [0.1198 \text{ in}^4]}{[48 \text{ in}]^2} = 14882.36 \text{ lbf}$$

$$\frac{P}{P_{cr}} = \frac{[8.6 \times 10^3 \text{ lbf}]}{[14882.36 \text{ lbf}]} = 0.5778$$

$$y_{max} = y_D = e \left[\sec \left(\frac{\pi}{2} \sqrt{\frac{P}{P_{cr}}} \right) - 1 \right] = [0.0625 \text{ in}] \left[\sec \left(\frac{\pi}{2} \sqrt{0.5778} \right) - 1 \right] = 0.10736 \text{ in} \quad \boxed{y_{max} = 0.10736 \text{ in}}$$

Max stress:

$$\sigma_{max} = \frac{P}{A} \left[1 + \frac{ec}{r^2} \sec \left(\frac{\pi}{2} \sqrt{\frac{P}{P_{cr}}} \right) \right] = \frac{[8.6 \times 10^3 \text{ lbf}]}{[1.22718 \text{ in}^2]} \left[1 + \frac{[0.0625 \text{ in}][0.625 \text{ in}]}{[0.3125 \text{ in}]^2} \sec \left(\frac{\pi}{2} \sqrt{0.5778} \right) \right]$$

$$\boxed{\sigma_{max} = 14626.65 \text{ psi}}$$

These results make sense since we superposed the bending moment due to the eccentricity and the normal stress on the beam. This shows that eccentricities cause a significant contribution to max stress in buckling compared to purely centric loading. [Time spent: 30 mins]